

Laboratory Simulation & Synthesis Report

Analysis of Basic Logic Gates and Combinational Half Adder Circuits

1. Introduction & VLSI Design Flow Overview

In modern Electronic Product Engineering and Very Large-Scale Integration (VLSI) design, a structured, hierarchical flow is vital to transform behavioral concepts into physical silicon implementations. The design flow generally spans several key stages:

1. System Specification

The initial stage where the chip's functionality, target application, and market requirements are defined. It establishes critical Engineering design constraints, including performance (clock speed), power consumption budgets, and silicon area limits (PPA targets).

2. Architecture Design

The high-level conceptualization phase. Engineers map out the macro-architecture of the system, defining major hardware blocks, memory interfaces, data pathways, and software-hardware boundaries using high-level block diagrams.

3. RTL Design

The transition into hardware description languages (HDLs). The architectural blocks are coded using languages like Verilog or VHDL to describe the **Register Transfer Level (RTL)** behaviour, specifying how data moves between registers and how it is transformed by combinational logic.

4. Functional Verification

A rigorous simulation phase dedicated to proving correctness. The RTL code is rigorously tested against the system specification using testbenches to ensure the logical behaviour is entirely bug-free before proceeding to physical implementation.

5. Synthesis

The translation phase where the validated, abstract RTL description is converted into a technology-dependent **gate-level netlist**. This process maps the HDL code onto actual logic gates (AND, OR, flip-flops) provided by a specific semiconductor foundry's standard cell library.

6. Physical Design

The core back-end engineering phase. It involves translating the gate-level netlist into a geometric layout. Key sub-stages include **floor planning** (allocating chip areas for blocks and I/O pads), **placement** (positioning standard cells), and **routing** (interconnecting the components with metal layers).

7. Timing & Power Analysis

A critical sign-off stage where static timing analysis (STA) and power estimation tools are used. This ensures that clock frequencies are met without setup or hold violations, and that the physical layout stays within the designated power limits under worst-case operating conditions.

8. Fabrication

The manufacturing phase. The finalized geometric layout (typically in GDSII or OASIS format) is sent to a semiconductor foundry (FAB), where photolithography and chemical processing steps physically print the circuits onto silicon wafers.

9. Testing & Validation

The final post-silicon phase. Once the manufactured physical ICs return from the foundry, they undergo rigorous hardware testing using automated test equipment (ATE) to filter out manufacturing defects and validate that the physical chip functions correctly in real-world environments.

2. Basic Logic Gates Simulation & Validation

Each foundational logic gate has been systematically modeled and simulated via Logisim. Below are the specific schematic diagrams, truth tables, and targeted observations.

2.1 Two-Input AND Gate

The AND gate implements logical multiplication where the output $Y = A \times B$ becomes high (1) if and only if both independent inputs are high.

Input A	Input B	Output Y (AND)
0	0	0
0	1	0
1	0	0
1	1	1

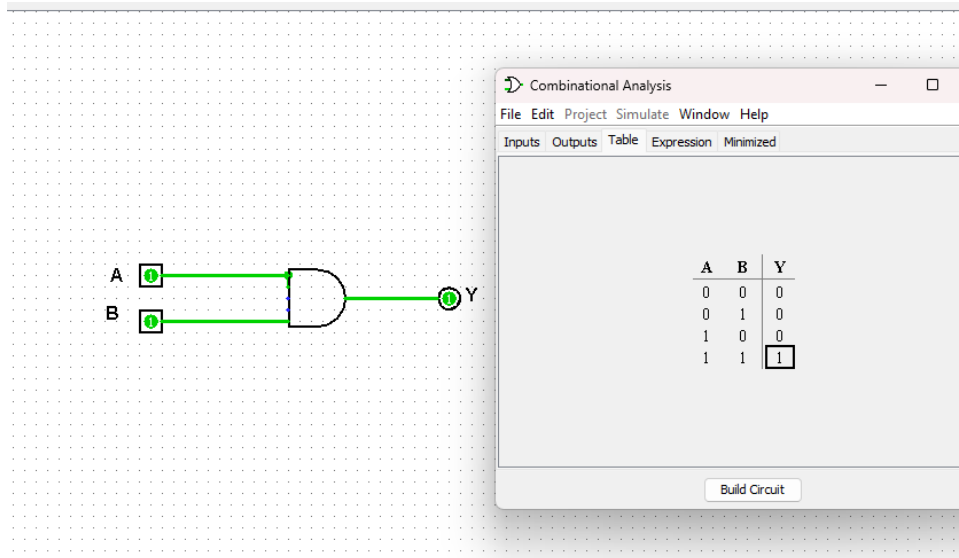


Figure 1: Schematic and verified truth table of the 2-Input AND gate

2.2 Two-Input OR Gate

The OR gate performs logical addition where the output $Y = A + B$ is pulled high whenever at least one input condition is asserted.

Input A	Input B	Output Y (OR)
0	0	0
0	1	1
1	0	1
1	1	1

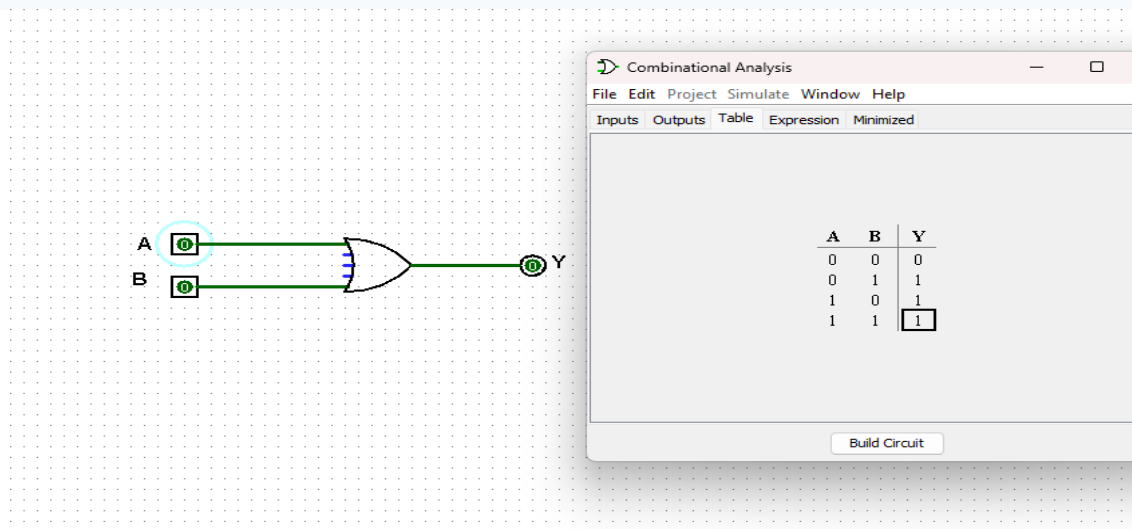


Figure 2: Schematic and verified truth table of the 2-Input OR gate

2.3 Inverter (NOT Gate)

The single-input NOT gate produces a complemented output signal ($Y = A'$), performing basic state inversion.

Input A	Output Y (NOT)
0	1
1	0

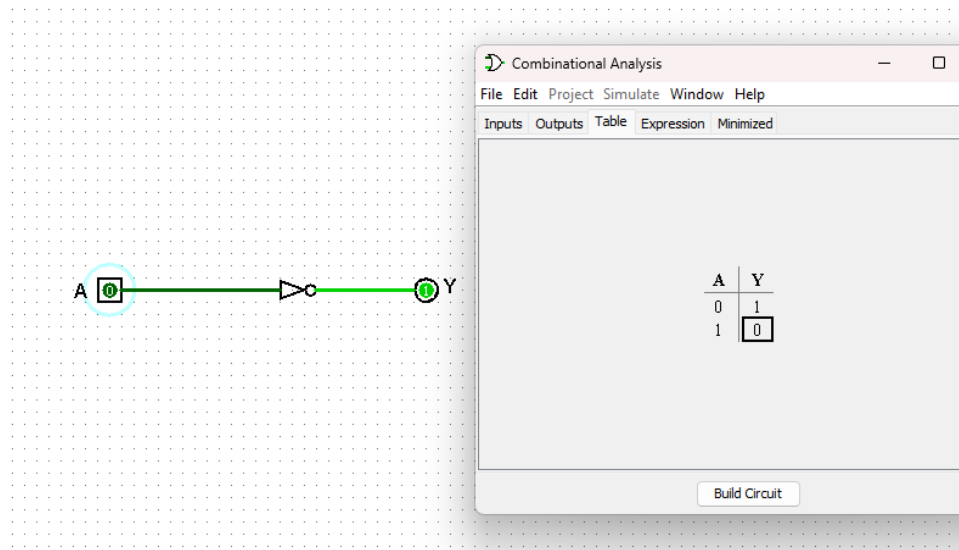


Figure 3: Schematic and verified truth table of the Inverter (NOT) gate

2.4 Universal NAND Gate

As a universal building block, the NAND gate produces a logic low output exclusively when all inputs are logically high.

Input A	Input B	Output Y (NAND)
0	0	1
0	1	1
1	0	1
1	1	0

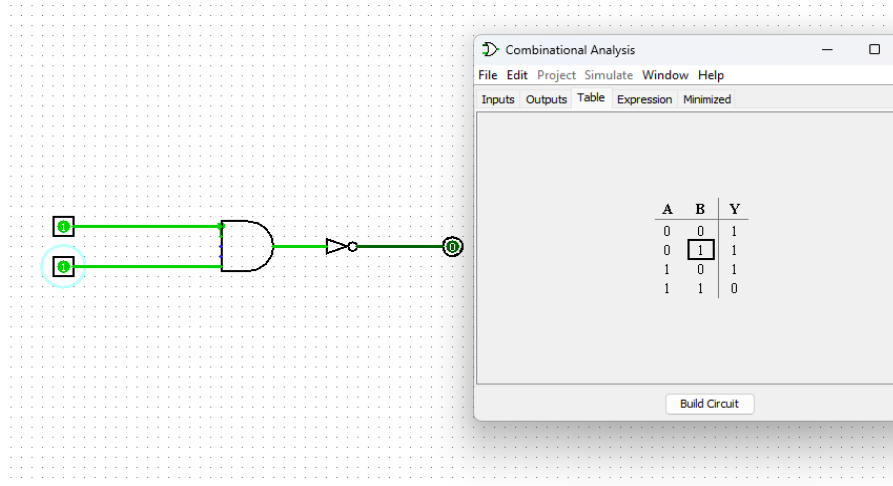


Figure 4: Structural layout and logic validation of the Universal NAND gate

2.5 Universal NOR Gate

The NOR gate represents a complemented OR function, producing a high output only when both input signals are inactive.

Input A	Input B	Output Y (NOR)
0	0	1
0	1	0
1	0	0
1	1	0

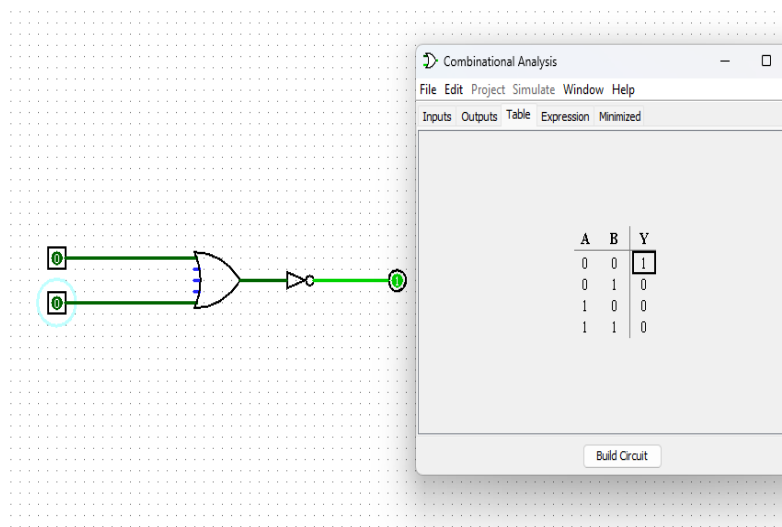


Figure 5: Structural layout and logic validation of the Universal NOR gate

2.6 Two-Input XOR Gate

The XOR (Exclusive-OR) gate outputs a logic high (1) if and only if the input states are different ($Y = A \oplus B$).

Input A	Input B	Output Y (XOR)
0	0	0
0	1	1
1	0	1
1	1	0

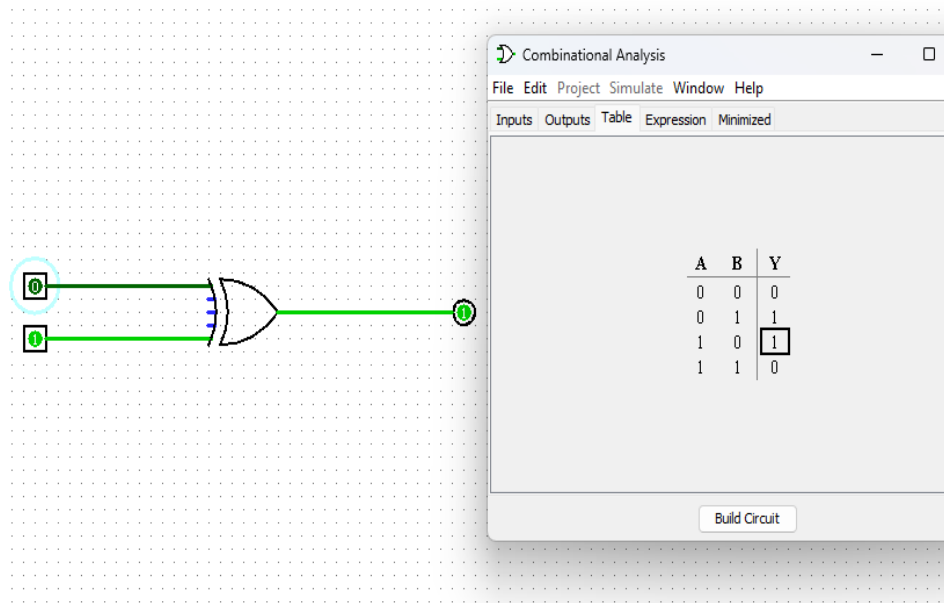


Figure 6: Structural layout and logic validation of the 2-Input XOR gate

3. Half Adder Circuit Synthesis & Simulation Analysis

The Half Adder is a foundational combinational subsystem capable of computing the arithmetic sum of two single-bit binary inputs, A and B. It yields two discrete functional outputs:

- SUM (S): Modeled by an Exclusive-OR (XOR) logic behavior where $S = A \oplus B$.
- CARRY (C): Modeled by an AND logic gate where $C = A \cdot B$.

3.1 Mathematical Truth Table

Input A	Input B	Output SUM (S)	Output CARRY (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

3.2 Structural Simulation View

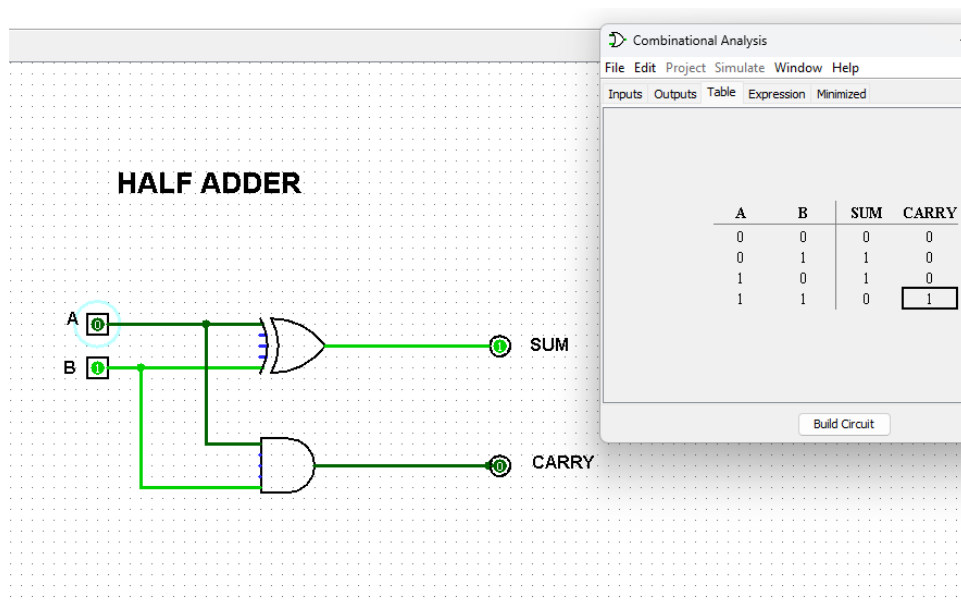


Figure 7: Complete schematic of Half Adder

The simulation results successfully corroborate the mathematical principles of Boolean algebra and digital logic design. Each elemental gate matched its exact expected functional response. In the synthesized Half Adder network, when both inputs are pulled high (A=1, B=1), the sum bit correctly toggles to 0 while generating a carry overflow bit of 1.

4. Conclusion

- Gained a comprehensive overview of the end-to-end industrial VLSI design flow, recognizing how a chip transitions from initial system specifications down to physical fabrication and testing.
- Reinforced that complex modern Integrated Circuits (ICs) and Systems-on-Chip (SoCs) are fundamentally built upon basic digital logic gates and Boolean algebraic principles.

- Acquired hands-on experience utilizing digital logic simulation tools (such as Logisim/Circuit Verse) to design, simulate, and debug circuits.
- Successfully implemented and verified all primary logic gates (AND, OR, NOT, NAND, NOR, XOR) alongside a functional combinational circuit (Half Adder).
- Confirmed that the hardware behaviour perfectly matches mathematical expectations by exhaustively testing all input combinations against known truth tables.